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*Article*

# Assessment of Published Papers on the Use of Machine Learning in Diagnosis and Treatment of Mastitis

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**Abstract:** The present study is an evaluation of published papers on machine learning as employed in mastitis research. The aim of this study was the quantitative evaluation of the scientific content and the bibliometric details of these papers. In total, 69 papers were found to combine machine learning in mastitis research and were considered in detail. There was a progressive yearly increase in published papers, which originated from 23 countries (mostly from China or the United States of America). Most original articles ($n$ = 59) referred to work involving cattle, relevant to mastitis in individual animals. Most articles described work related to the development and diagnosis of the infection. Fewer articles described work on the antibiotic resistance of pathogens isolated from cases of mastitis and on the treatment of the infection. In most studies (98.5% of published papers), supervised machine learning models were employed. Most frequently, decision trees and support vector machines were employed in the studies described. 'Machine learning' and 'mastitis' were the most frequently used keywords. The papers were published in 39 journals, with most frequent publications in *Computers and Electronics in Agriculture* and *Journal of Dairy Science*. The median number of cited references in the papers was 39 (interquartile range: 31). There were 435 co-authors in the papers (mean: 6.2 per paper, median: 5, min.–max.: 1–93) and 356 individual authors. The median number of citations received by the papers was 4 (min.–max.: 0–70). Most papers (72.5%) were published in open-access mode. This study summarized the characteristics of papers on mastitis and artificial intelligence. Future studies could explore using these methodologies at farm level, and extending them to other animal species, while unsupervised learning techniques might also prove to be useful.

**Keywords:** algorithm; artificial intelligence; cattle; machine learning; mammary infection; mastitis; meta-research; scientometric analysis; staphylococcus; subclinical mastitis

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## 1. Introduction

### 1.1. Background: Mastitis

Mastitis is defined as the inflammation of the mammary gland. The infection is caused mainly by bacteria invading the mammary parenchyma. The causal bacteria are contagious and include *Staphylococcus aureus*, non-*aureus* staphylococci (with *S. chromogenes*, *S. epidermidis* and *S. simulans* as the predominant species), *Streptococcus agalactiae*, *Mycoplama bovis*, *Corynebacterium bovis* [1,2], environmental pathogens, such as *Strep. uberis*, *Escherichia coli* and other coliforms (e.g., *Klebsiella* spp., *Enterobacter* spp.) and *Pseudomonas* spp. [2–4]. Other pathogens, involved less often in the etiology of the infection, include various fungi (e.g., *Aspergillus* spp., *Candida* spp.) [5], algae (e.g., *Prototheca* spp.) [6] or viruses (e.g.,

bovine herpesvirus 1, bovine herpesvirus 4, foot-and-mouth disease virus, Parainfluenza 3 virus) [7].

Mastitis is characterized by a high leucocyte content in milk and in the mammary parenchyma, along with shedding of the causal pathogens in milk. In this respect, it is noted that, in some cases, intramammary infections in cattle can be nevertheless manifested with short duration or intermittent bacterial shedding [8]. Based on the clinical symptoms, mastitis is classified into (i) clinical mastitis, characterized by the presence of clinically evident abnormalities and (ii) subclinical mastitis, which can be diagnosed only based on the results of laboratory examinations. Clinical features include systemic signs (e.g., increased body temperature) and signs localized in the affected mammary gland. Among the latter, enlargement of the affected mammary gland, discoloration of the udder skin, development of nodules or hardness within the mammary gland occur more often; further, there may be changes in the mammary secretion (clotted or hemorrhagic). The main laboratory findings include the recovery of pathogens from milk and the increased leucocyte numbers therein ('high somatic cell counts'). The infection adversely affects the welfare of affected mammalian species. Clinical mastitis is a painful disorder, which leads to pathophysiological and behavioral changes in affected animals [9].

Moreover, in dairy animals, the infection has a paramount financial importance, as its primary consequences relate to reduced milk production and the downgrading of milk quality. Milk is a valuable agricultural commodity globally and is ranked among the top five agricultural commodities worldwide [10]. The financial impact of the infection includes (a) the cost of losing the affected animals that die, (b) the cost of premature culling of affected animals, (c) the cost of purchasing replacement animals, (d) the cost of reduced milk production by affected animals, (e) the cost of reduced purchase price of milk of inferior quality, (f) the cost of veterinary services and drugs, (g) the cost of discarding milk unfit for human consumption and (h) the cost of increased labor in farms. The cost of a case of mastitis has been reported to be around EUR 280, leading to an average cost of approximately EUR 30 per animal on a farm [11], although variations occur between countries. Overall, the global cost of bovine mastitis is claimed to vary between EUR 16 and 30 billion annually [12,13].

The complexity of the disease and the significant financial losses have led to devising and developing a variety of diagnostic approaches for the infection, which offer quick, accurate and confirmatory diagnosis [14]. In general, diagnostic tests of intramammary infections can be divided into general, which detect general mastitis indicators, changes and deviations irrespective of the causal pathogens [15–17], and specific, which detect specific pathogens causing the infection, their genetic material, estimate biomarkers or show changes related to specific pathogens [18,19].

With regard to the treatment of mastitis, that is more frequently performed by relying on antibiotic administration [20], while new published studies describe the development of various relevant therapeutic approaches [21]. The wide dissemination of antibiotic-resistant bacteria causing mastitis has reiterated the need for careful and prudent use of these antimicrobials in the treatment of the infection [22].

### *1.2. Background: Machine Learning*

Machine learning (ML) is part of artificial intelligence and refers to the development and the use of algorithms, which may learn from data and generalize to unseen data, thereby performing tasks without explicit instructions [23,24]. Currently, generative artificial neural networks have surpassed other, previously employed, approaches in performance [25]. The mathematical foundations of machine learning are provided by optimization methodologies [26]. Machine learning approaches have been applied to various scientific fields and everyday life aspects, including agriculture and medicine, where it is too costly to develop algorithms to perform the needed tasks [27–29].

Computational statistics is an important component in the relevant methodologies employed in these methodologies. Approaches employed in machine learning include

'supervised learning', in which inputs and desired outputs are presented and made available for model training and algorithm construction, and 'unsupervised learning', in which the learning algorithm is left to find structure in the input, with no labels given [30,31].

### *1.3. Background: Use of Machine Learning in Mastitis Research*

The rationale of studies employing machine learning methodologies to mastitis research is to improve the diagnosis and treatment of the infection, given that these methods can 'learn' from data patterns and may exploit information available in datasets. The availability of automated monitoring systems in dairy cattle farms supports the collection of such data.

Details of various parameters, such as the volume of milk produced by cows, milk flow and the milking time of individual animals, the protein and lactose concentration in milk and the electrical conductivity of milk, are monitored and collected, thus providing large amounts of data for setting up and developing models relevant to mastitis. It is noteworthy that various studies have employed large datasets for setting up models for mastitis diagnosis. For example, Ebrahimie et al. [32] used 345,000 milking records and Pakrashi et al. [33] used records of 1,350,000 milk-days from 2390 cows in their studies applying machine-learning methodologies in mastitis.

Mastitis is a multifactorial infection involving a significant interplay between the animal, the pathogen and environment. Hence, the application of machine learning methodologies can thus help veterinarians to make clinical decisions regarding the diagnosis of the disease and to craft effective treatment and prevention plans.

### *1.4. Objectives of This Study*

The present study is an evaluation of published papers on machine learning, as employed in mastitis research. The aim of this study was the quantitative evaluation of the scientific content and the bibliometric details of these papers. This study addresses the quantity and the impact of information published internationally on using machine learning in mastitis research. This article can be of use primarily by researchers on mastitis and on computer applications in agriculture; further, it can provide information to veterinarians, animal scientists, computer professionals and data analysis workers.

Bibliometric studies are similar to systematic reviews in the retrieval of the relevant literature [34], but they have different objectives than these. There is also a low rate of agreement between the two types of studies with regard to the retrieved literature. Systematic reviews aim to respond to a pre-defined question based on the available evidence. In contrast, bibliometric–scientometric analyses give emphasis to the analytical assessment of the available evidence, without quality assessment [35].

## 2. Materials and Methods

Papers combining the fields of mastitis and machine learning were identified through the Web of Science platform. These papers were analyzed with regard to their scientific content and their bibliometric details, in order to provide a detailed scientometric analysis on the topics under study.

### *2.1. Search Procedure*

The Web of Science platform (www.webofknowledge.com; Clarivate Analytics, London, UK) was used for the search of relevant publications. For the search, we used the Web of Science Core Collection, in a search that spanned across multiple disciplines; the platform includes the Emerging Sources Citation Index, the Science Citation Index Expanded, the Arts and Humanities Citation Index, the Social Sciences Citation Index, the Book Citation Index and the Conference Proceedings Citation Index.

A topic search using the following terms was carried out: [mastitis OR *mammary infection*] AND [artificial intelligence OR [machine learning OR machine-learning] OR deep learning OR decision tree* OR vector machine* OR naive Bayes OR k-nn OR neuronic

OR anomaly detection OR association rules OR recommendation systems OR algorithm* OR architecture* OR optimization]. In this string, the asterisk served as a truncation symbol to include variations of the terms. A topic search retrieved records that included the query terms in the title, in the keywords or in the abstract. The search was performed on 8 February 2024 ('freeze date'). Only records published up to the 31 December 2023 were included in this study.

The initial search produced 520 records. Thereafter, the document analysis of these records was performed, during which the following types of documents were only included: 'article' and 'review article'. Thus, 515 papers were retained for further assessment individually.

### *2.2. Paper Evaluation*

Evaluation of qualitative characteristics in the papers was performed by two pairs of evaluators: one pair assessed mastitis characteristics and one pair machine learning characteristics. The evaluators worked independently and compared their results after completion of their work. In case of disagreement between the two assessors in each pair, the paper was evaluated by a senior author (author: G.C.F.). This occurred with two papers in relation to the assessment of mastitis characteristics. In these cases, the qualitative characteristics of the paper were recorded as those agreed upon by two of the three evaluators.

During evaluation of these papers, those not including work related to mastitis or to artificial intelligence (including machine learning) were excluded from further evaluation. Finally, after the above, 69 papers remained and were included in the scientometric evaluation. In each of these papers, the following details were recorded:

- Year of publication of paper.
- Country and scientific establishment (university or other institution) of origin of the paper were taken into account based on the first and last authors only. If multiple authors were listed as first or last authors in the papers, they were all considered.
- Type of paper: (i) original article or (ii) review. For original articles, the following details were further recorded:
- Mammalian species involved in the study described.
- Focus of study described in the paper: (i) individual patient, (ii) population.
- Mastitis aspect(s) described therein: (i) diagnosis, (ii) treatment.
- Type of study described therein: (i) experimental work (i.e., challenge-associated), (ii) field work or (iii) laboratory work.
- Specific feature associated with mastitis studied in the article: (i) presence of clinical mastitis and/or subclinical mastitis, (ii) presence of chronic mammary abnormalities, (iii) presence of bacteria in milk, (iv) increased somatic cell counts in milk, (v) findings of clinical pathology examination, (vi) findings of histopathology examination, (vii) outcome of mastitis treatment and (viii) isolation of antibiotic-resistant bacteria from milk.
- Methodological approaches related to machine learning employed in the study described.
- Journal in which the paper was published.
- Number of literature references included in the relevant list.
- Number and names of all co-authors in the paper.
- Total number of citations received by the paper until the end of 2023.
- Accessibility of paper, i.e., whether it was open access or subscription-only.

### *2.3. Data Management and Analysis*

The number of papers published on [mastitis OR *mammary infection*] only was also assessed using the same procedure, in order to compare with the number of papers published on the topic of the current study.

To assess the impact of the papers published, the total number of citations received by the papers was employed. The number of citations received by the papers was normalized for the year of publication by calculating the average citations per year since publication for each paper.

All data were entered into Microsoft Excel (v. 2401-2406). Descriptive analysis was performed initially. The frequency of the various outcomes was evaluated in tables of cross-categorized frequency data using the Pearson chi-square test or the Fisher exact test, as appropriate. Comparisons of proportions were performed by a two-proportion z-test. Comparisons between continuous data were performed using the Mann–Whitney test or Kruskal–Wallis test. Linear regression analysis was used to establish associations with the year of publication of each paper. Spearman rank correlation analysis was performed as indicated, and the significance of the result was evaluated according to the critical values for $r$.

Statistical significance was defined at $p < 0.05$.

## 3. Results

All the 69 papers individually assessed were indexed in the Web of Science, fulfilled the search criteria and presented work on mastitis by using machine learning (Table S1).

### *3.1. Year of Publication*

There was a clear increase in the number of papers published progressively: slope 1.12 ± 0.24 ($p < 0.0001$) (Figure 1). The increase in the number of papers published since 2017 was significantly higher than that of all papers published on mastitis: slope 2.43 ± 0.72 versus 77.00 ± 24.29, respectively ($p = 0.012$) (Figure 2, Table S2). There was a significant increase in the proportion of papers on mastitis and artificial intelligence among all mastitis papers from 2017 to 2023 (slope: 1.78 ± 0.62; $p = 0.017$).

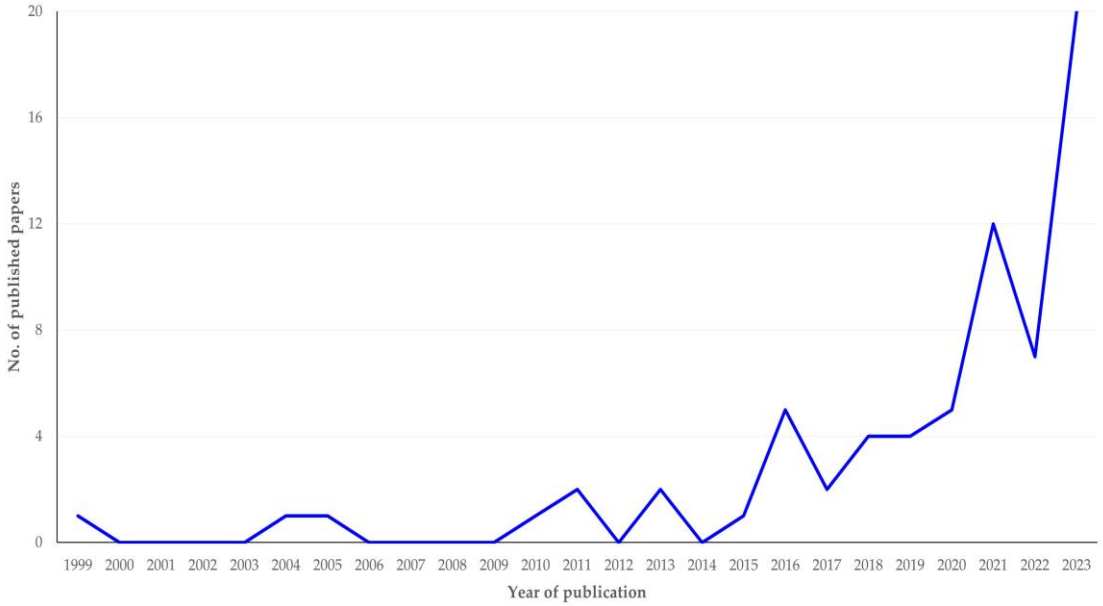



**Figure 1.** Graph of the number of published papers on mastitis and artificial intelligence according to the year of publication.

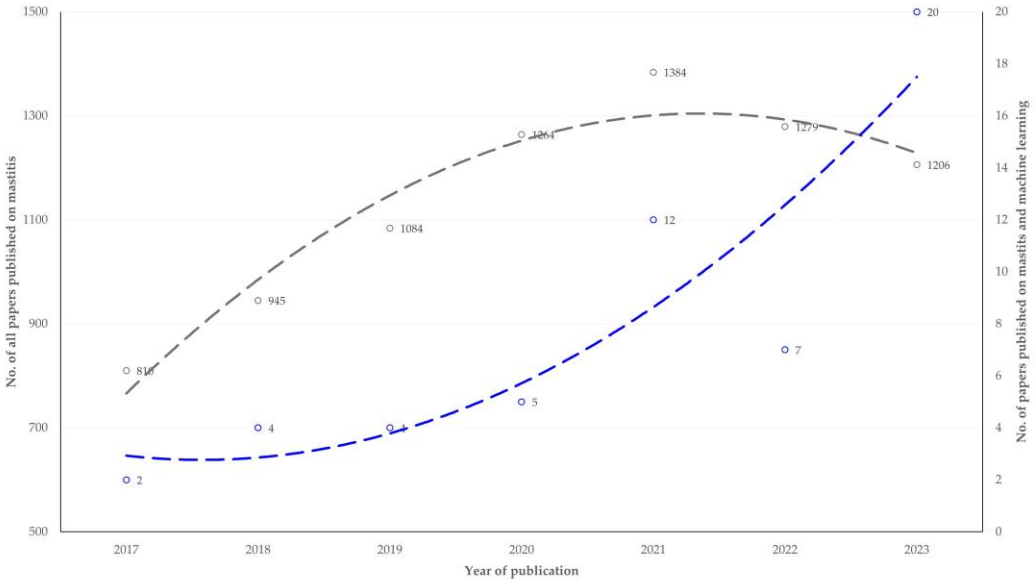



**Figure 2.** Trendlines for number of published papers on mastitis (grey) or on mastitis and artificial intelligence (blue) during the period from 2017 to 2023 (figures within the graph are actual data).

### *3.2. Origin of Papers*

The papers originated from a total of 23 countries and 70 scientific establishments. The median number (interquartile range) of published papers per country was 2 (3); most papers originated from China ($n$ = 13) and the United States of America ($n$ = 10) (Figure 3, Tables S3 and S4). The median number of published papers per scientific establishment was 1 (1); most papers originated from China Agricultural University and University of Nottingham ($n$ = 5 each) and from Inner Mongolia Agricultural University and University of Adelaide ($n$ = 4 each) (Table S5).

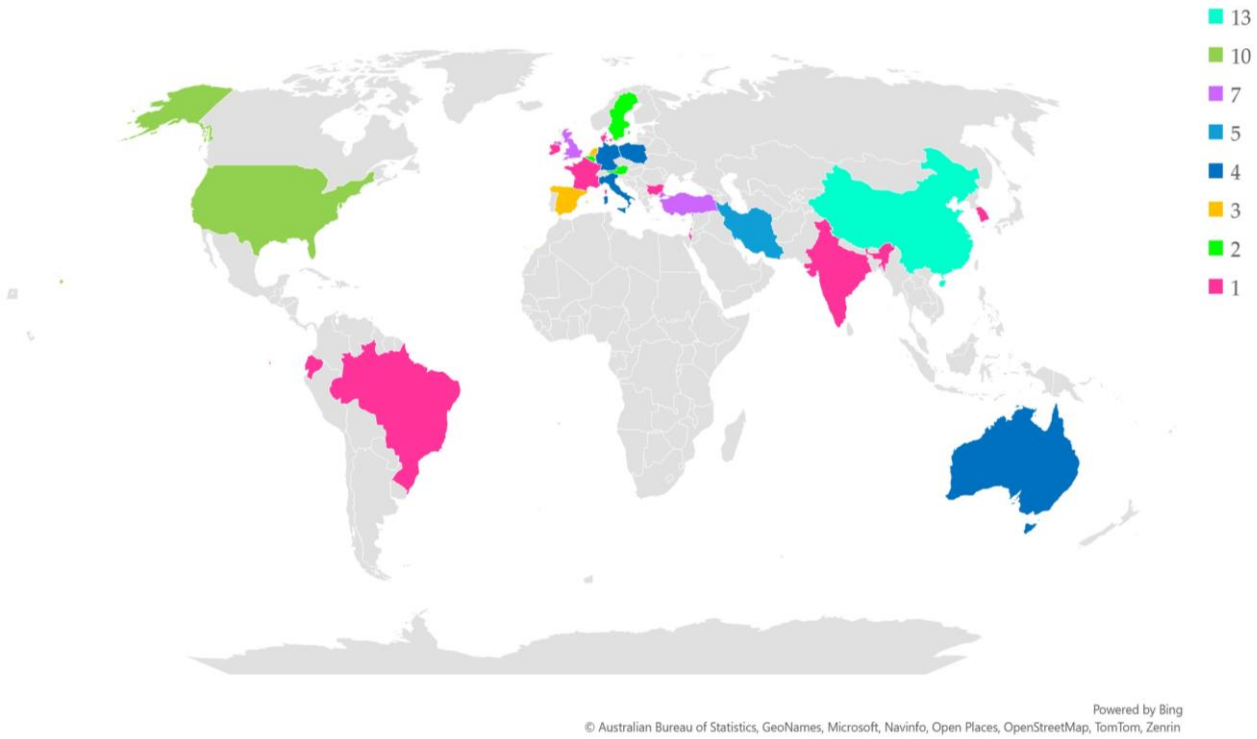



**Figure 3.** Countries and respective number of papers on mastitis and artificial intelligence that have originated from each.

The scientific establishments from which the relevant papers originated included 56 universities and 14 establishments of other type (e.g., research centers, research departments of animal health companies, hospitals). However, there was no difference between the two types of establishments in the median number of papers that originated from each: 1 (1) versus 1 (0), respectively ($p = 0.20$).

There was a significant difference in the median year of publication of the papers among the 22 countries ($p = 0.013$) (Table S3); Belgium, Brazil, China, France, Ireland and Italy had the most recent median year of publication (2023) for the papers that originated from these countries. A progressive increase in the number of countries and scientific establishments from which published papers originated was noted. Until the end of 2019, papers originated from 14 countries and 31 scientific establishments; thereafter, papers originated from 18 countries and 42 scientific establishments cumulatively.

### *3.3. Content of Papers*

Most of the papers were original articles ($n = 65$, 94.2%), while there were also four reviews (5.8%).

#### 3.3.1. Mammalian Species Referred to in Articles

The large majority of original articles presented work performed on cattle ($n = 58$; 89.2%). Fewer original articles presented work performed on humans ($n = 3$), buffaloes ($n = 3$) or pigs ($n = 1$) (Figure 4).

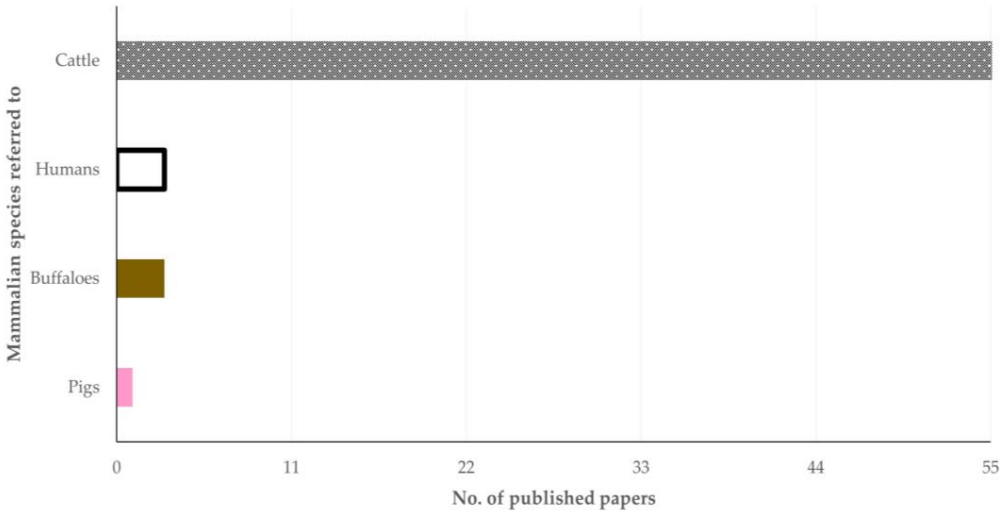



**Figure 4.** Number of original articles on mastitis and artificial intelligence, in accordance with the mammalian species referred to in the respective studies.

#### 3.3.2. Mastitis Aspect Described in Articles

All original articles ($n = 65$, 100.0%) presented work referring to mastitis in individual patients, i.e., at cow, woman or sow level. Moreover, one article ($n = 1$, 1.5%) also described work on mastitis in cattle at farm level.

The majority of articles described work relevant to the diagnosis of mastitis ($n = 59$; 90.8%). These studies most frequently referred to work for the identification of clinical mastitis ($n = 20$), the detection of increased somatic cell counts in milk ($n = 12$) or the diagnosis of clinical and subclinical mastitis ($n = 12$). Fewer articles described work relevant to the treatment of the infection ($n = 9$; 13.8%) (Figure 5, Table S6).

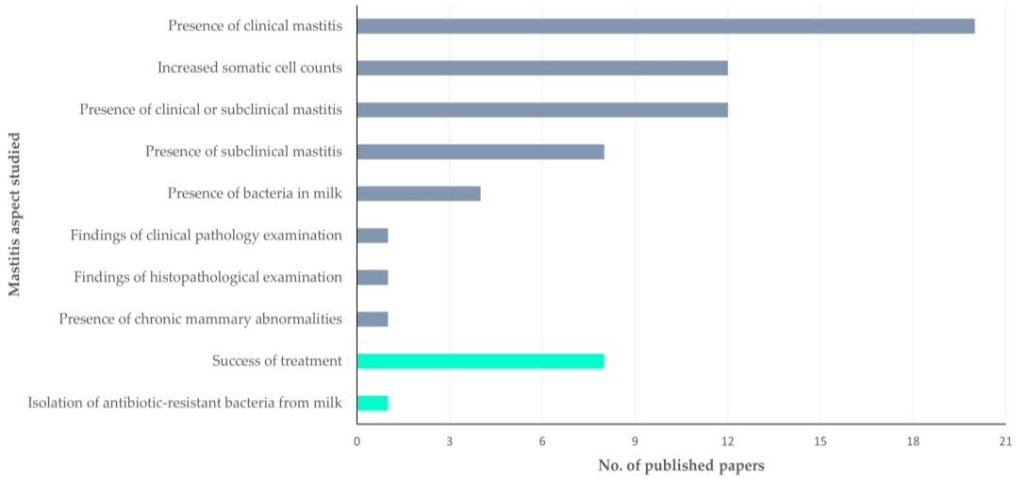



**Figure 5.** Number of original articles on mastitis and artificial intelligence, in accordance with the mastitis aspect studied therein: diagnosis (blue grey) or treatment (green) of the infection.

With regard to the type of mastitis in the 54 papers on the diagnosis of the infection, most ($n$ = 20) described the diagnosis of clinical mastitis only (18 in cattle, 1 in humans and 1 in pigs); fewer papers ($n$ = 8) dealt with the diagnosis of subclinical mastitis only (all in cattle), while 12 papers described the diagnosis of both clinical and subclinical mastitis (cattle); finally, one paper dealt with the diagnosis of 'chronic' mastitis (cattle). There was no significant difference in the median year of publication of the articles with regard to the mastitis aspect studied therein ($p > 0.14$ for all comparisons) (Table S7). In addition, there was no significant difference in the proportion of papers that originated from each country with regard to the mastitis aspect studied therein ($p = 0.89$) (Table S8).

Among the papers that described applications in the treatment of mastitis, all referred to the treatment of clinical mastitis (seven papers regarding mastitis during the lactation period and one during the dry period). The pathogens implicated in the mastitis cases included in these studies represented a wide array of mastitis-causing bacteria (*S. aureus*, coagulase-negative staphylococci, *Strep. agalactiae*, *Strep. dysgalactiae*, *Strep. uberis*, *E. coli*, *Klebsiella* spp.). The median accuracy of predicting treatment success using machine learning methodologies was 83% among the papers.

#### 3.3.3. Machine Learning Methodologies Employed

In the vast majority of studies (in 64 of the 65 original articles, 98.5%), supervised machine learning models were employed. Unsupervised methodologies were described in one article (1.5%). Eight different methodologies were employed in the studies described (Table 1). The methodologies employed most frequently were decision trees ($n$ = 41, 68.3%) and support vector machines ($n$ = 17, 28.3%).

**Table 1.** Machine learning methodologies employed in studies described in original articles on mastitis and artificial intelligence, along with the number of papers describing the use of each.

| Machine Learning Methodology | Number of Original Articles ($n$ = 60) |
|---|---|
| Decision trees | 41 (68.3%) |
| Support vector machines | 17 (28.3%) |
| Artificial neural networks | 16 (26.7%) |
| Logistic regression | 9 (15.0%) |
| k-nearest neighbors algorithm | 7 (11.7%) |
| Naïve Bayes classifiers | 6 (10.0%) |
| Generalized linear models | 4 (6.7%) |
| Linear regression | 4 (5.0%) |
| k-means | 1 (1.7%) |
| (not mentioned) | 6 |

The median number of methodologies employed was 1 (1) per original article. The most frequently employed combinations of methodologies were (i) decision trees and logistic regression, (ii) decision trees, support vector machines and artificial neural networks and (iii) decision trees, support vector machines, k-nearest neighbors algorithm and logistic regression. The use of each of these combinations was described in two (2) articles.

There was no difference in the median number of methodologies employed between articles describing the diagnosis or treatment of mastitis: 1 (1) versus 1 (3), respectively ($p$ = 0.24). However, there was a difference in the type of methodologies employed in accordance with the mastitis aspect described in the paper. In studies on the treatment of mastitis, decision trees and logistic regression were described more frequently than in the studies on the diagnosis of the infection: 100.0% and 66.0% ($p$ = 0.037) compared to 44.4% and 9.4% of the respective papers ($p$ = 0.006), respectively (Table S9). However, there was no difference in the median year of publication of articles describing the use of the various different methodologies ($p$ = 0.77) (Table S10).

#### 3.3.4. Detection of Antibiotic Resistance in Isolates from Cases of Mastitis by Using Machine Learning Techniques

In one paper, the objective of the study was to find alternative methods to standard tests for detecting antibiotic-resistant isolates from cases of mastitis. The study used supervised learning methodologies, specifically decision trees, support vector machines, Naïve Bayes classifiers, k-nearest neighbors algorithm, logistic regression and artificial neural networks. The methods were employed in *S. aureus* isolates ($n$ = 82) recovered from cases of mastitis in cattle in the United Kingdom, with the objective to detect multidrug resistance (15 isolates) or resistance to benzylpenicillin (16 isolates). Overall, the accuracy of detection of resistant isolates was found to be 97% and 98%, respectively. Support vector machines were found to be the machine learning technique through which the highest accuracy could be obtained.

#### 3.3.5. Keywords

In total, 187 different keywords were included in the 69 papers; the median number of keywords per papers was 5 (1). Of these, 37 keywords were included in at least two papers: 'mastitis' (in 22 papers), 'machine learning' (in 21 papers each), 'dairy cow*' (in 9 papers) and 'prediction' and 'somatic cell count*'(in 5 papers each) were included more frequently (Table S11).

In total, 588 different pairs of keywords were found. Of these, 566 were unique pairs, whilst 22 pairs of keywords were found in more than one paper. The pairs of keywords detected more frequently were 'machine learning' and 'mastitis' ($n$ = 15) and 'machine learning' and 'dairy cow*' ($n$ = 5) (Figure 6, Table S12).

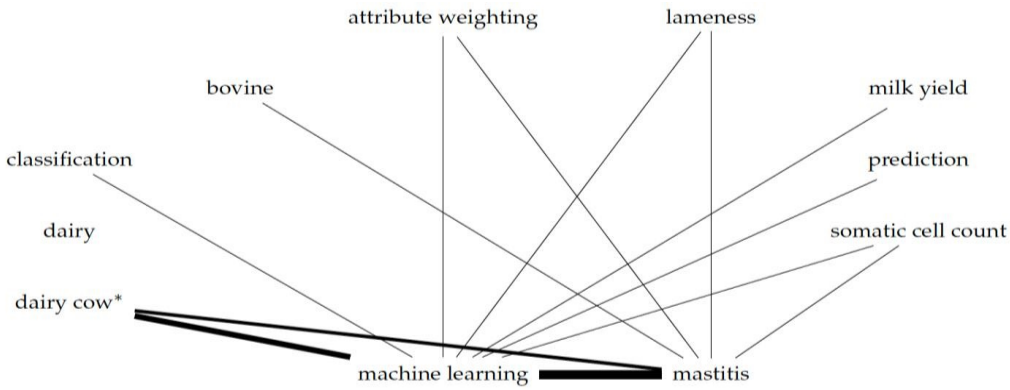



**Figure 6.** Pairs of keywords found in more than one published paper on mastitis and artificial intelligence (with the width of lines joining keywords corresponding to the number of respective published papers) (dairy cow*: * is used as a truncation symbol).

### 3.4. Journals in Which Papers Were Published

The papers were published in 39 different journals (Table S13). The journals in which most papers were published, were the following: *Computers and Electronics in Agriculture* ($n$ = 10), *Journal of Dairy Science* ($n$ = 9), *Animals* ($n$ = 7) and *Scientific Reports* ($n$ = 4). Overall, 43.5% of the papers were published in these four journals.

It was noted that the papers originating from the Netherlands were published exclusively in *Computers and Electronics in Agriculture* ($n$ = 3), the papers from the United States were published mainly in *Journal of Dairy Science* ($n$ = 3) or in *Computers and Electronics in Agriculture* ($n$ = 2) and the papers from the United Kingdom were published frequently in *Scientific Reports* ($n$ = 3). Moreover, the papers from China were published frequently in *Animals* or *Computers and Electronics in Agriculture*, the papers from Germany in *Sensors* and the papers from Italy in *Animals* ($n$ = 2 in each). Further, three articles that employed more than one methodology in the studies described were published in both *Computers and Electronics in Agriculture* and *Journal of Dairy Science*. Two such articles were published in both *Scientific Reports* and *Sensors*.

The sub-categories in the Web of Science, in which journals with published papers were classified, are shown in Table S14. The sub-categories with most papers were *Agriculture, Dairy and Animal Science* ($n$ = 23 papers), *Veterinary Sciences* ($n$ = 29 papers), *Agriculture, Multidisciplinary* ($n$ = 13 papers), Food Science and Technology ($n$ = 12 papers) and *Computer Science, Interdisciplinary Applications* ($n$ = 11 papers).

### 3.5. Cited References

The median number of cited references per published paper was 39 (interquartile range: 31). Original articles included a significantly lower number of cited references than reviews: 38 (22) versus 109.5 (20.5), respectively ($p$ = 0.003) (Figure 7).

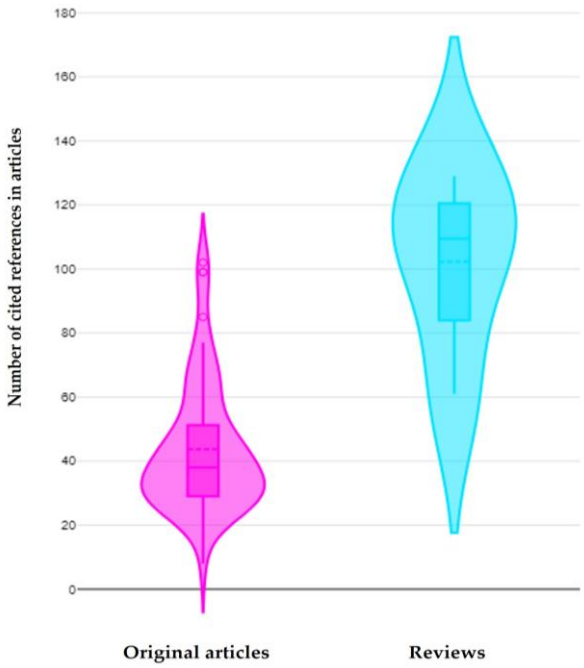



**Figure 7.** Violin plot of the number of cited references in original articles (pink) or reviews (turquoise) on mastitis and artificial intelligence.

There was a clear positive correlation between the year of publication of the paper and the number of cited references therein ($r_{sp}$ = 0.337, $p$ = 0.005). Moreover, the published papers presenting work on mastitis treatment had a significantly higher number of cited references than the papers presenting work on mastitis diagnosis: 50 (32) versus 36 (19.5), respectively ($p$ = 0.007) (Figure 8).

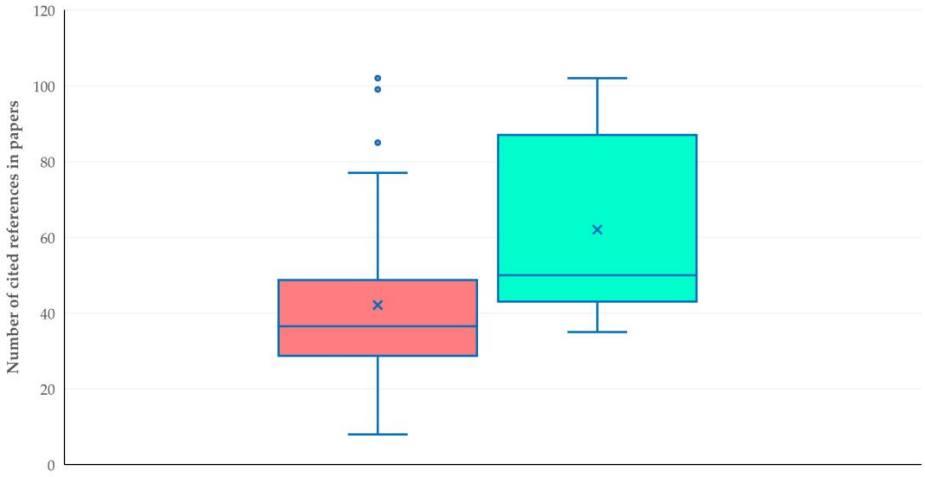



**Figure 8.** Box and whisker plot of the number of cited references in original articles on mastitis and artificial intelligence, presenting work on diagnosis (red) or treatment (bright green) of the infection.

### *3.6. Authors*

There were in total 356 individual authors of the papers. Cumulatively, in the 69 papers, there were 435 co-authors, i.e., on average 6.2 ± 1.1 co-authors per paper (median: 5 (2), min.–max.: 1–93). There were 14 authors with at least three papers (max.: 6); of these, three were among the 50 authors with most published papers on mastitis alone. There was only one (1.4%) single-author paper.

Further, there were 118 individual authors who were first or last authors in the papers. Among these, 15 authors were first or last in at least two papers each (max.: 3). These 15 authors were first or last authors in 90.4% ± 4.7% of their papers (min.: 40.0%, max.: 100.0%). All these authors were affiliated with scientific establishments in the 10 countries with most papers published; three of these authors showed affiliation with two different scientific establishments. Six pairs of authors, affiliated concurrently with the same establishment in each of six different countries, were identified with common papers.

There was a difference among the 11 countries with the most published papers in the median number of authors per paper published ($p = 0.049$) (Table S15). Further, no significant differences were identified in the number of authors among papers that referred to studies on diagnosis or to studies on the treatment of mastitis: 5 (2) versus 6 (3), respectively ($p = 0.38$), nor were there differences in accordance with the number of methodologies employed in the relevant studies ($r_{sp} = 0.015$, $p = 0.91$).

The average number of co-authors per paper increased gradually throughout the years (slope: 0.28 ± 0.14; $p = 0.048$) (Figure 9). Finally, the median number of authors in original articles did not differ significantly to that in reviews: 5 (2) versus 4.5 (1.5), respectively ($p = 0.90$).

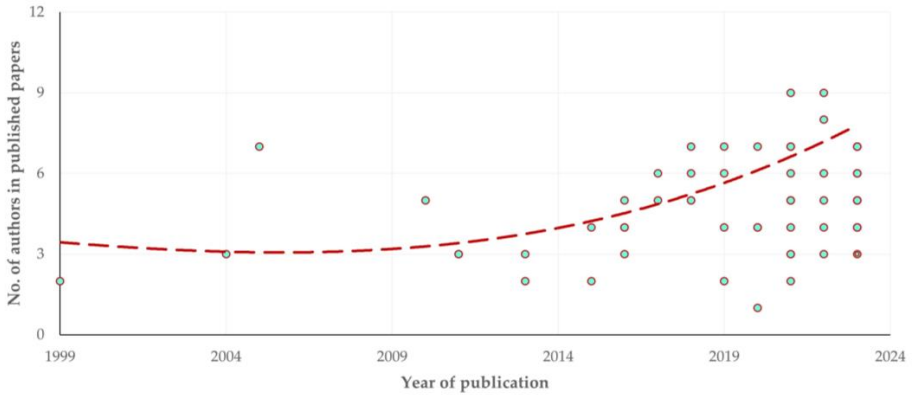



**Figure 9.** Number of co-authors in papers on mastitis and artificial intelligence, in accordance with the year of publication (dashed line is trendline).

### *3.7. Impact of Papers*

The median number of citations received by the 69 papers was 4 (interquartile range: 13) (min.–max.: 0–70) and the median yearly number of citations was 2.0 (3.6) (min.–max.: 0.0–14.9).

There was no difference in the median number of yearly citations received by reviews or original articles: 1 (3) versus 2 (3.4), respectively ($p$ = 0.44). Additionally, there was no correlation between the yearly number of citations and the number of cited references ($r_{sp}$ = −0.017, $p$ = 0.89) or the number of authors ($r_{sp}$ = 0.145, $p$ = 0.24). With regard to the content of the paper, there was no significant association for yearly number of citations between papers that presented studies on the diagnosis or treatment of mastitis: two (3.6) versus two (1.8), respectively ($p$ = 0.47). In addition, there was no correlation with the number of machine learning methodologies employed in the study ($r_{sp}$ = −0.00.26, $p$ = 0.84).

There were, however, no differences in the number of yearly citations of the published papers according to their country of origin ($p$ = 0.033). The countries whose papers received most yearly citations were Australia and Iran, with the median numbers of yearly citations being 9.5 (6.0) and 6.7 (8.2) per paper, respectively (Table S16).

### *3.8. Accessibility of Papers*

Of the 69 published papers, most ($n$ = 50, 72.5%) were published under open-access and fewer under subscription-only access ($n$ = 19, 27.5%). The median year of publication of the former papers was significantly more recent than that of the latter ones: 2021 (3) versus 2018 (5.5) ($p$ = 0.007) (Figure 10).

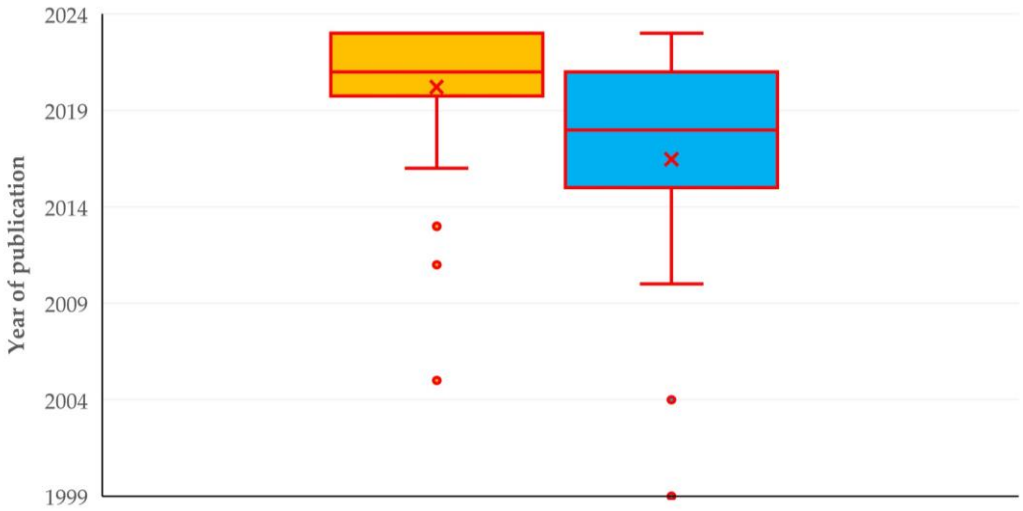



**Figure 10.** Box and whisker plot for the year of publication of published papers on mastitis and artificial intelligence, published under open-access (orange) or subscription-only access (blue).

The proportion of papers submitted for open-access publication was highest among the papers from Italy (100%) and Poland (75.0%) ($p$ = 0.64 between countries). Finally, the number of yearly citations received by published papers did not differ in accordance with the accessibility of papers: 2.0 (3.4) versus 1.6 (3.2) ($p$ = 0.81), for papers published under open- or subscription-only access.

## 4. Discussion

Scientometric studies can help to summarize with accuracy and reliability the available evidence [36]. Indeed, scientometric evaluation and the use of relevant indicators can be useful tools in the assessment of research on a given topic [36]. This approach can help to grasp the available literature, identify knowledge gaps, produce new ideas and pathways

for further studies and, more importantly, position the potential contributions of these investigations in the wider context of the topic [37].

Scientometric studies combine quantitative and descriptive characteristics and may also make pronouncements about the qualitative aspects [35]. Through their use, one may 'translate' the scientific quality into manageable entities [38]. These studies rely on and benefit from the interpretation of quantitative details of publications. These include the topics of research within a field, the authors and the research groups, the scientific collaborations and interactions, the sources of publication, the origin (countries and scientific establishments) of papers published and the impact of the articles within the mapped literature [35]. Through scientometric studies and bibliometric analyses, it is possible to note the trends that emerge in the publication of papers, the performance of scientific journals and the preference of researchers for publishing in some of these. In addition, these studies reveal the patterns of collaboration between research groups and explore and uncover the structure of the existing literature in the study field [35].

### *4.1. Year of Publication*

Although the term 'machine learning' was coined in 1959 [39], its use in the life sciences has surged during recent years [25,27]. In machine learning, large amounts of data can be processed to uncover informative patterns and relationships within the data, even with limited prior knowledge of the system in question [40]. Therefore, there has been a rapid growth in the development of software designed for the analysis and interpretation of data using machine learning methodologies. These encompass an extensive array of machine learning algorithms, which can be utilized for various tasks, e.g., clustering, classification, regression, feature selection [41].

The above trends have also become evident in the papers evaluated during the present study, most of which were published during the last four years. In fact, the number of relevant papers has increased at a faster rate than the number of all papers on mastitis alone, indicating that researchers have been using the new technology to improve knowledge on mammary health. The higher number of papers published in recent years, coupled with the increasing number of countries and scientific establishments from which the relevant papers originated, reflects the interest of scientists in extending the use and the application of these new methodologies.

### *4.2. Content*

#### 4.2.1. Mastitis

In these articles, the primary objective was to discern patterns and relationships within the data, enabling the creation of predictive models relevant to mastitis studies. Most of the original articles identified and assessed in the present study reported research in which machine learning applications were employed to achieve predictions regarding diagnostic outcomes in individuals, mostly in cattle. Indeed, an automated tool with increased accuracy in mastitis diagnosis can be useful for veterinary clinicians. Cow milk accounts for 80% to 85% of the global milk production [10], so most relevant studies were performed on cattle.

In general, the respective studies used algorithms to successfully predict the aimed outcomes based on previous data obtained from the same individuals. There is merit in this approach, as mastitis can recur in some animals as the result of genetic, management or environmental influences [42,43]. That way, the occurrence of mastitis in a cow can be predicted and therefore appropriate measures can be taken to minimize the risk and, potentially, prevent the infection.

Information employed by machine learning systems included mostly data obtained from records available on farms, e.g., from information obtained during milking or from health records of individual animals. Such data were used to make predictions regarding the development of mastitis, with accuracies that varied from 76% to 94% [44–46]. The accuracy of prediction increased nearer the time of the predicted event, specifically, one

day prior to the forecasted event for subclinical mastitis, with 80% accuracy [33], or 2 days prior to the forecasted event for clinical mastitis, with 84% accuracy [47]. This interval can be useful for implementing measures in the meanwhile that could ultimately prevent the occurrence of the adverse event in an animal.

Another type of information employed for the detection of clinical mastitis involved video images of the udder of cows. In such cases, the models employed were based on vision-based automatic recognition of abnormal features (size and temperature) of the udder of cows. The accuracy of this approach was found to be as high as 87% [48], as it combines the detection of the increased size of the udder with the higher temperature of the organ, obtained by means of infrared thermography, which are evident in clinical mastitis.

Nevertheless, the lack of relevant studies on mastitis at population level is noted, e.g., studies on the prevalence of the infection within a farm. Such studies would have been useful in the assessment of control measures to be applied at population level, e.g., the evaluation of potential problems with the milking system, the necessity for anti-mastitis vaccination.

Moreover, the lack of studies on small ruminant mastitis is also noted. A possible reason for this can be the reduced mechanization of sheep and goat farms, in comparison to cattle farms. Lianou et al. [49] recently reported that in Greece machine-milking was applied on only 72% of sheep and goat farms and among these, only 1% had automated monitoring systems in the milking parlors. Obviously, these conditions make it impossible to collect data during the milking of the animals on the farms.

Fewer articles referred to assessing the potential outcome of the treatment of mastitis. Complete and effective treatment is important for the cessation of bacterial shedding on the farm (by means of which pathogens disseminate to other animals within a population) and for the return of affected animals to full milk production [2,50]. The presence of antibiotic resistance among mastitis-causing pathogens is a significant problem; thus, predicting the isolation of such bacteria can be particularly useful in formulating therapeutic strategies.

During the treatment of intramammary infections, with regard to the 'One Health' perspective, clinicians should assess the balance between the potential risk of intramammary infections adversely affecting the health, welfare and production of affected animals and the need for the reduction in use of antimicrobial agents, which contributes to reducing antibiotic resistance. The early instigation of the treatment of infected animals leads to high cure rates of the animals and thus to more rapid restoration of milk production [51–53]. Moreover, this approach helps avoid the development of chronic mammary lesions, which are a primary reason for culling dairy ruminants [53,54].

The best approach for accurately identifying infected animals for treatment is the microbiological examination of mammary secretion samples, which nevertheless requires time and can be costly [52,53,55]. The practical advantage of applying protocols developed and established through the use of machine learning is that they allow for the effective use of antibiotics, thus decreasing the risk of emergence of antibiotic resistance.

#### 4.2.2. Machine Learning

In most studies, the methodology employed was decision trees, while other methodologies were utilized much less frequently. This was rather surprising, as decision trees is a rule-based method and one of the earliest used methods in machine learning, while other methodologies applied in the field of machine learning have not gained as much popularity in this context. This may possibly be a consequence of mastitis being a multifactorial disease [56,57], the diagnosis and treatment of which require a combination of approaches [57–59]. The predictions related to the diagnosis of the infection entail a stochastic process, in which the appropriate models are formulated to establish relationships between features and outcomes. This enables professionals to make informed decisions regarding the potential development of the infection, followed by action plans about its control. Therefore, it is common to utilize computational and statistical machine learning methods for diagnosis.

The models developed for use in mastitis research and clinical work must be explainable and carefully validated and interpreted. Moreover, ensuring the explainability and transparency of these models is important for garnering trust from researchers and clinicians. This aspect is inherently present in decision trees, which can justify the popularity of the methodology. However, transparency may be lacking in computational-statistical machine learning methods, in which the issue of the 'black box' remains a concern.

It is noted that machine learning includes a diverse array of methodologies that can be applied to data related to mastitis. These include, among others, supervised, unsupervised, semi-supervised learning, reinforcement, deep learning, anomaly detection, recommendation systems, association rules, etc. However, the observations reveal that, within the field of mastitis research, supervised learning methodologies have predominantly been utilized. Despite the rich landscape of machine learning techniques available, supervised learning approaches have been the primary focus in the investigation of mastitis-related data.

The preference of researchers to use primarily two methods, specifically decision trees and support vector machines, may be related to the advantages offered by these. Decision trees have an intuitive and easy-to-understand structure. Further, they may be converted into a set of human-readable rules, allowing for the straightforward interpretation of the decision-making process. In addition, the method does not make any assumptions regarding the distribution of data or the relationship between the variables in the dataset, which sets it apart from many other machine learning algorithms [60]. Support vector machines are effective in handling high-dimensional data and can find complex decision boundaries, resulting in accurate predictions. Additionally, the method is less prone to overfitting compared to other algorithms, e.g., neural networks. Support vector machines cannot be as easily interpretable as decision trees. However, they can identify points closest to the decision boundary and can offer insights into the model's decision-making process; this can aid in understanding the importance of different features in the classification process [61]. Overall, the combination of these advantages makes the two methods prominent choices for application in machine learning works, which can explain their increased usage in research studies.

Moreover, various other available methodologies have not been applied at all in mastitis work. Among them, anomaly detection can be employed in data mining and machine learning in order to identify patterns deviating from the norm or expected behavior within a dataset. Reinforcement learning is a method of machine learning where an agent learns to make decisions by interacting with an environment. In contrast, recommendation systems are a method using algorithms designed to predict and suggest items of interest to users based on their preferences or past interactions. As such, these systems are commonly used in e-commerce, streaming platforms and social media to personalize user experiences by offering relevant content, products or services [62,63].

Nevertheless, dairy farmers may now employ machine learning methodologies in the control of mastitis. These can be a 'game-changer', as they will make it possible (i) to diagnose mastitis at the early stages, which will allow for the earlier intervention and instigation of treatment; (ii) to predict cows with a higher chance of developing the infection, which will support focusing the resources on a farm for preventing the infection is such animals; and (iii) to better allocate resources, as with better information regarding the risk for mastitis development, time and money will not be spent on unnecessary interventions, but rather on strategies that can be targeted to animals in greater need for these.

A search that included the search terms [artificial intelligence OR machine learning OR machine-learning OR deep learning OR decision tree* OR vector machine* OR naive Bayes OR k-nn OR neuronic OR anomaly detection OR association rules OR recommendation systems] AND [neoplasia OR tumour OR tumor OR cancer OR malignan*] AND [lung] revealed 6200 papers published until the end of 2023, i.e., a figure almost 10,000% higher than the number found in the field of mastitis. This huge difference can explain why the papers assessed in the current study refer only to supervised learning algorithms rather than to more complex systems. Likely, the limited extent of relevant research and,

consequently, the fewer opportunities for development hinder attempts to develop and use more complex methodologies and models. Indeed, a recommendation systems algorithm requires a complicated methodology for development; thus, it may not be desirable to invest in development within a rather limited field. Hence, this can be a limiting factor in using more complex methodologies in mastitis research.

The conditions on livestock farms may also contribute to the low numbers of machine learning models applied in mastitis (or other topics in veterinary and animal science). The education level of farmers and labourers may not be as high as that of hospital personnel (e.g., nurse, laboratory staff, physiotherapists). In the European Union, specifically, 72% of farmers had not received full agricultural training [64]. This difference may contribute to limiting the potential applications of machine learning methodologies on farms and explain the difference in published papers between the agricultural sector and the medical world, as indicated above, which in turn hinders investments in machine learning methodologies.

#### 4.2.3. Potential Limitations

Despite the use of a search string of 14 terms, the extended array of artificial intelligence methodologies and techniques could have resulted in some published papers being missed. It is noted that, in the rapidly evolving field of machine learning, new methodologies and tools are continuously developed and emerge for practical application [65], potentially surpassing the techniques taken into account and reviewed herein. Further, other authors could have used different keywords and terminology in their papers, or just synonyms of the terms applied in the current search, which would have also resulted in omitting these papers.

Moreover, some reports might have been published outside the established academic publishing channels indexed in the Web of Science database. These may include conference proceedings, theses and dissertations, technical reports (i.e., the so-termed 'grey literature' [66]), which were thus omitted. On the other hand, however, the use of the WoS database, which only accepts journals for inclusion if some quality criteria were met (e.g., periodicity of publication, reviewing process etc.), confirms some minimum standards of quality of the journals. Nevertheless, as is customary among scientists, many of these publications will subsequently be published in peer-reviewed journals. For example, the preliminary presentations in scientific conferences are often submitted to journals as full manuscripts for potential publication. Thus, these papers will become available when updating, in the future, the list of papers within the scope of this study.

Inconsistencies in conducting the studies and variations in the reporting of their findings might also have contributed to potentially omitting some papers from this study. In order to keep this to a minimum, four assessors evaluated each paper for inclusion, with a senior author available to resolve possible conflicts between them.

Despite the above, this paper offers a valuable overview of the use of artificial intelligence and machine learning in mastitis work and summarizes the currently available studies, thus providing a reference for the future.

### *4.3. Bibliometric Details*

The Web of Science platform was used as the source for the records, given that it is a bibliometric platform unaffiliated to any publishing house, unlike other similar platforms. Moreover, the Web of Science offers stricter quality control measures for the journals included in their database.

Many papers were published in journals with a specific focus on computer sciences (e.g., *Computers and Electronics in Agriculture*, *Computers in Biology and Medicine*, *International Journal of Pattern Recognition and Artificial Intelligence*) or journals with a thematic focus on animal studies (e.g., *Journal of Dairy Science*, *Animals*, *Frontiers in Veterinary Science*). In contrast, few papers were published in multidisciplinary journals (e.g., *Scientific Reports*, *Applied Sciences-Basel*, *PlosOne*), despite the topic being clearly a multidisciplinary one.

Hence, thematic journals seemed to be more appealing to researchers for publication of research output to specialized readers with similar scientific interests.

Open-access publishing increases continuously, as seen by the increasing proportion of papers published under this model, in compliance with the terms for the dissemination of results of publicly funded research [67,68]. Indeed, the dissemination of scientific knowledge and scientific communication is promoted by open-access publishing, which helps to make the results of scientific research available easily and free of charge to scientists and the wider public [69].

## 5. Conclusions

This study summarized the characteristics of papers on mastitis and artificial intelligence—machine learning. In general, the relevant literature is limited, which indicates significant possibilities for fruitful research on the topic. The methodologies employed were focused on the diagnosis and the treatment (including the detection of antibiotic resistant pathogens) of the infection in cattle and at individual animal level. Supervised learning methodologies were employed. Machine learning algorithms played an important role in predicting cow mastitis. It has become evident that relevant research can support clinical decisions.

Future studies might explore additional aspects of mastitis, including the development and progression of the infection. Further, the methodologies should be employed at farm level, in order to establish the frequency of the infection. This will be of particular importance in the health management of populations of dairy animals. There is also a clear potential to apply these methodologies in sheep and goats, in which species mastitis is also a significant disease, affecting their welfare and reducing milk production. As it was noted that no studies applied unsupervised learning techniques, these techniques can also be explored to assess their potential usefulness for supporting clinical decisions by veterinarians.

Knowledge gained through machine learning methodologies can benefit in various sectors within dairy farm management: (i) early detection and diagnosis of mastitis through pattern recognition and the development of predictive models, (ii) optimization of mastitis treatment through outcome prediction and antibiotic resistance patterns, (iii) improvement of milk quality management through the analysis of somatic cell counts in milk.

Within this context, future possibilities can include the following: (i) incorporation of smart sensors and IoT (e.g., temperature sensors and/or activity trackers, which may continuously collect data for subsequent analysis through machine learning methodologies to provide real-time insights and alerts.), (ii) the instalment of robotic milking systems (for monitoring udder health, detecting early signs of mastitis and adjusting milking routines to minimize stress and infection risk, through the application of machine learning methodologies) and (iii) the development of mobile applications (e.g., applications for smartphones and tablets that can provide access to machine learning-powered insights and recommendations, enabling them to monitor animal and herd health and to manage mastitis effectively).

Finally, familiarization of those involved in farm work with the relevant technology will be necessary for the correct implementation of these technologies in practical conditions and for the correct evaluation of their outputs. Moreover, the collection of data by farmers and staff during daily routines will contribute to improving the algorithms used in developing models for diagnosis and treatment of mastitis and will improve the on-farm use and application of these models.

**Supplementary Materials:** The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/info15080428/s1, Table S1: Details of 69 papers published until 31st December 2023 on mastitis and machine learning and listed on the Web of Science platform; Table S2: Number of published papers on mastitis or on mastitis and artificial intelligence during the period from 2017 to 2023; Table S3: Countries and respective number of papers on mastitis and artificial intelligence that originated from each; Table S4: Alphabetic list of countries and respective number of

published papers on (a) mastitis and (b) mastitis and 17 artificial intelligence that originated from each; Table S5: Scientific establishments and respective number of papers on mastitis and artificial intelligence that originated from each; Table S6: Specific features of mastitis described in original articles on mastitis and artificial intelligence, along with the number of papers addressing each feature; Table S7: Median year (interquartile range) of publication of original articles on mastitis and artificial intelligence, in accordance with the specific features of mastitis in the respective studies; Table S8: Numbers of original articles on mastitis and artificial intelligence, in accordance with the specific features of mastitis described therein and the country from which the papers originated; Table S9: Machine learning methodologies employed in studies described in original articles on mastitis and artificial intelligence, in accordance with the mastitis aspect in the respective studies; Table S10: Median year (interquartile range) of publication of original articles on mastitis and artificial intelligence, in accordance with the machine learning methodology employed in the respective studies; Table S11: Keywords in original articles on mastitis and artificial intelligence and respective number of papers with each of these; Table S12: Pairs of keywords found in more than one papers on mastitis and artificial intelligence and the respective number of papers addressing each feature; Table S13: Journals in which papers on mastitis and artificial intelligence were published and respective number of papers; Table S14: Sub-categories of journals in the Web of Science, in which papers on mastitis and artificial intelligence were published, and respective number of papers; Table S15: Median (interquartile range) number of authors per published paper on mastitis and artificial intelligence, in accord with countries of origin of the papers; Table S16: Median (interquartile range) number of yearly citations received by published papers on mastitis and artificial intelligence, in accordance with the countries of origin of the papers.

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